

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

Instructor: Ijaz Ul Haq, Ph.D.

ijazhaq@umich.edu

Location: MSB 104



Lecture 07
Classification
(Linear Models, Model Evaluation)

Previous class:

- **Naive Bayes**
  - ***Where to use it?***
  - ***How it works?***
- **k-Nearest Neighbor (k-NN)**
  - ***Where to use it?***
  - ***How it works?***
- **Related NoteBook (code)**

Previous class:

- ***Logistic Regression Model***
  - ***Where to use it?***
  - ***How it works?***
- ***Support Vector Machines (SVM)***
  - ***Where to use it?***
  - ***How it works?***
- ***Model Evolution***

Classification: Logistic Regression

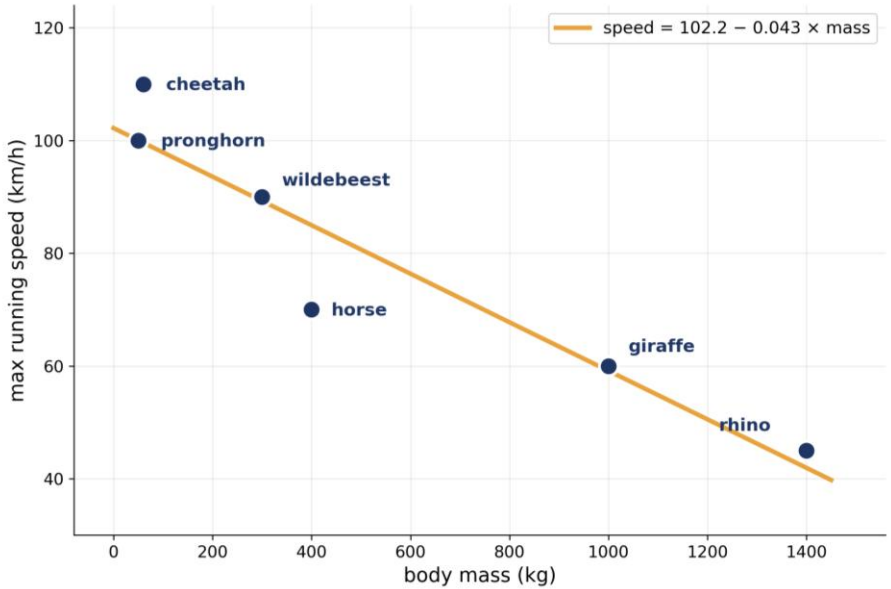
What is Regression?

Regression predicts a numeric value from a set of input features by fitting a function to the data.

Chapter 4
Introduction to Data Mining, 2nd Edition

Regression tries to determine how a dependent variable and one or more other (independent) variables relate to each other. It's a statistical method.

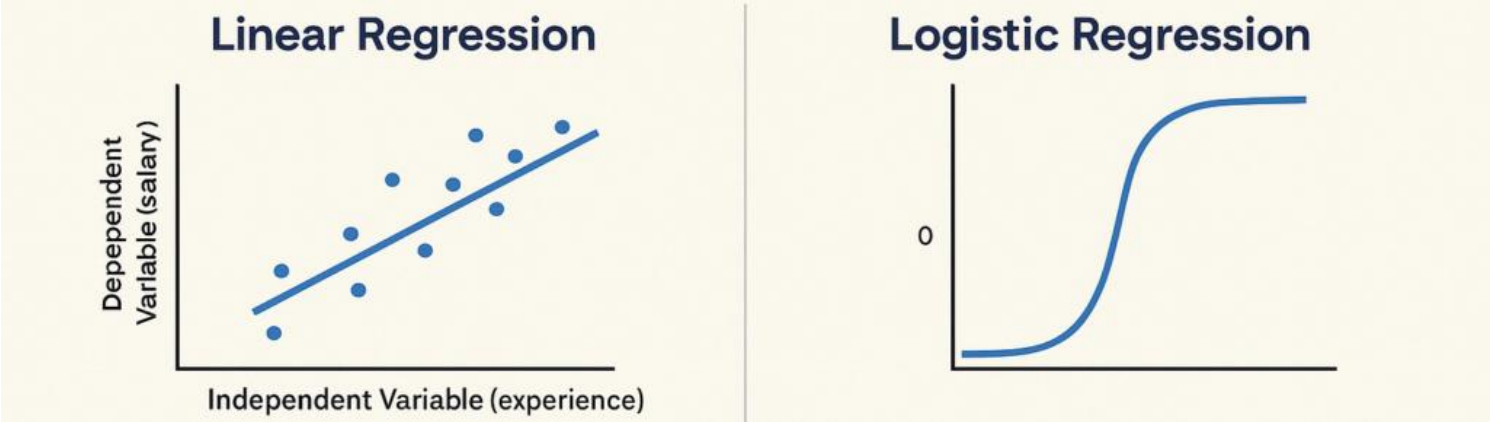
Animal	body mass (kg)	max running speed (km/h)
rhino	1400	45
horse	400	70
pronghorn	50	100
giraffe	1000	60
wildebeest	300	90
cheetah	60	110



Classification: Logistic Regression

Different Types of Regression:

- Linear regression
- Logistic regression



	Linear Regression	Logistic Regression
Predicts	A number	A class
Task	Regression	Classification
Output range	Anything, minus infinity to plus infinity	Always between 0 and 1
What the output means	The predicted value	The probability of belonging to a class
Shape of the fit	A straight line	An S curve

Classification: Logistic Regression

Logistic Regression:

Logistic regression models the probability that an object belongs to a class, as a function of its attributes, by fitting a linear function to the log-odds of that probability.

Chapter 4
Introduction to Data Mining, 2nd Edition

Where to use ?

- Credit scoring, where the lender must explain the decision
- Medical risk, where you need a probability rather than a yes or no
- Marketing response, will this customer convert

Pros

- Gives you a probability, not just a label
- Fast and stable on large data

Cons

- Needs scaling for regularization to be fair across features
- Struggles with imbalance unless you weight the classes

Classification: Logistic Regression

The Working of Logistic Regression

Given how many hours a student studied, will they pass or fail?

Independent dependent

Study Hours	Exam Results
2	Fail
5	Pass
4	Pass
1	Fail
6	Pass
3	Fail

Classification: Logistic Regression

The Working of Logistic Regression

Given how many hours a student studied, will they pass or fail?

Independent dependent

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**?**

0=Fail
1=Pass

… probability that an object belongs to a class…

What is the probability of a student that studied for 3.3 hours?

Classification: Logistic Regression

The Working of Logistic Regression

Given how many hours a student studied, will they pass or fail?

Independent dependent

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**?**

0=Fail
1=Pass

… probability that an object belongs to a class…

What is the probability of a student that studied for 3.3 hours?

A value between 0 and 1

Classification: Logistic Regression

The Working of Logistic Regression

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**?**

0=Fail
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Classification: Logistic Regression

The Working of Logistic Regression

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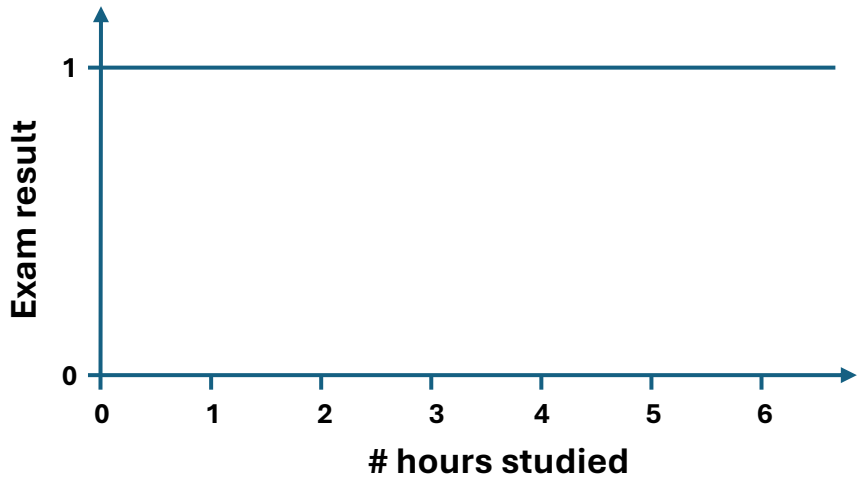


Classification: Logistic Regression

The Working of Logistic Regression

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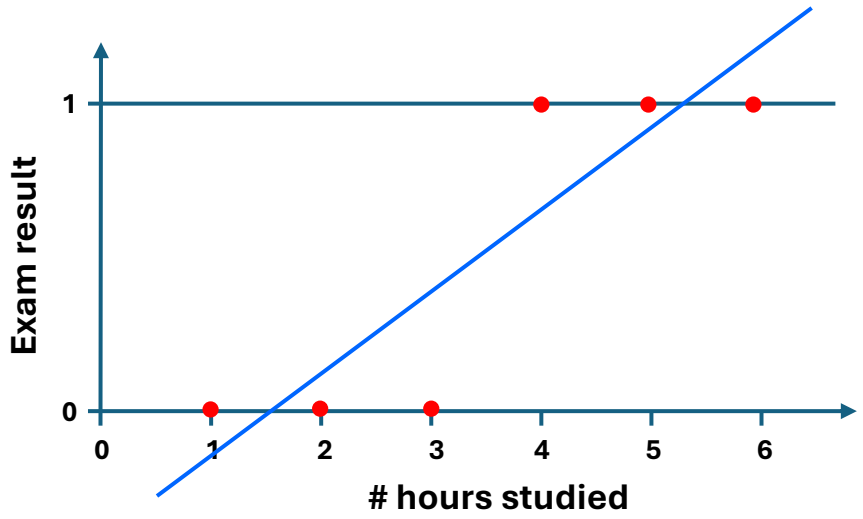


Classification: Logistic Regression

The Working of Logistic Regression

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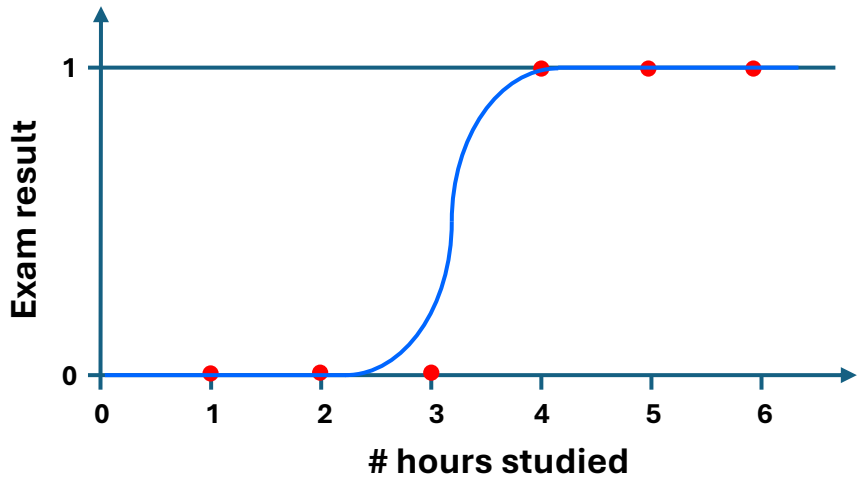


Classification: Logistic Regression

The Working of Logistic Regression

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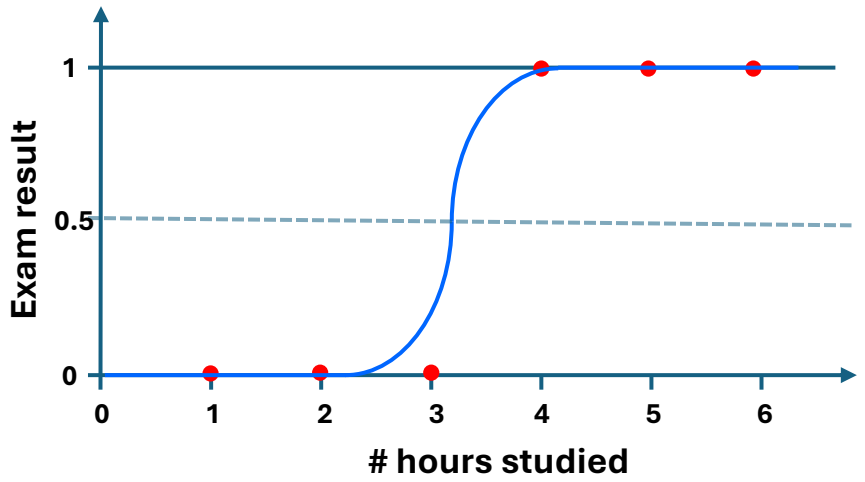


Classification: Logistic Regression

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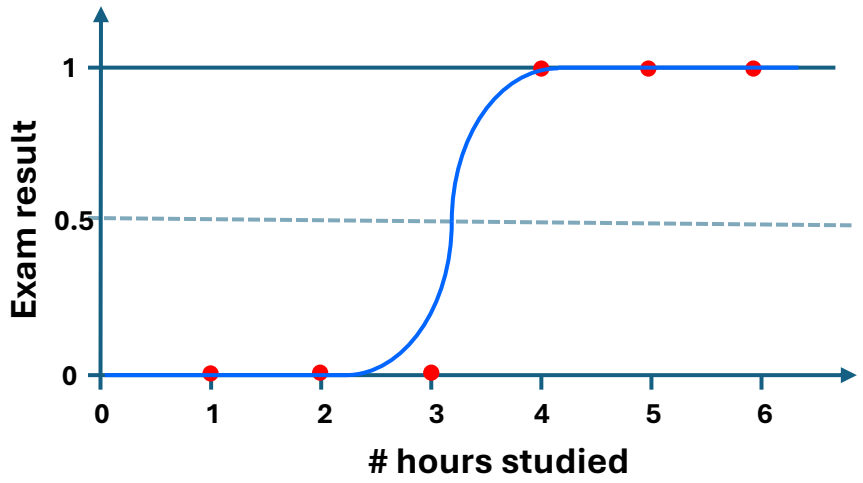
Classification: Logistic Regression

The Working of Logistic Regression

Study Hours	Exam Results
2	0
5	1
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6	1
3	0
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0=Fail
1=Pass

$$y = \frac{1}{1+e^{-(a_0+a_1x)}}$$



Classification: Logistic Regression

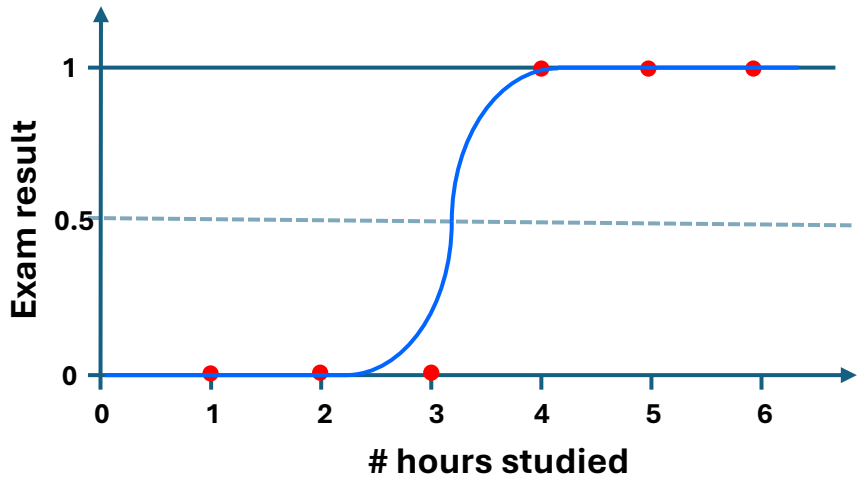
The Working of Logistic Regression

$$y = \frac{1}{1+e^{-(a_0+a_1x)}}$$

y = predicted probability (between 0 and 1)
a_o = intercept (probability when x=0)
a_1 = coefficient (how strongly x pushes y up or down)
X = independent variable

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**?**

0=Fail
1=Pass



Classification: Logistic Regression

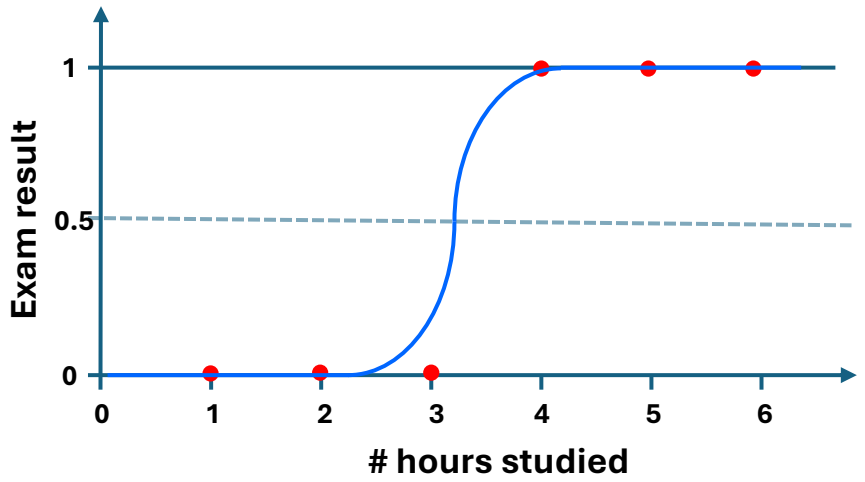
The Working of Logistic Regression

$$y = \frac{1}{1+e^{-(a_0+a_1x)}}$$

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**?**

0=Fail
1=Pass

y = ?
a_o = -1.5
a_1 = 0.6
X = 3.3



Classification: Logistic Regression

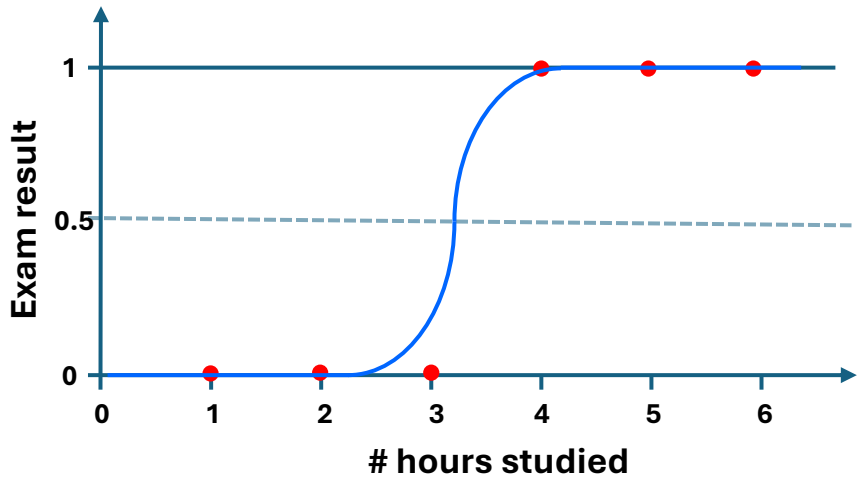
The Working of Logistic Regression

$$y = \frac{1}{1+e^{-(a_0+a_1x)}}$$

$$y = \frac{1}{1+e^{-(-1.5+0.6 \text{ x } 3.3)}}$$

y = ?
a_o = -1.5
a_1 = 0.6
X = 3.3

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**?**



0=Fail
1=Pass

Classification: Logistic Regression

The Working of Logistic Regression

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**?**

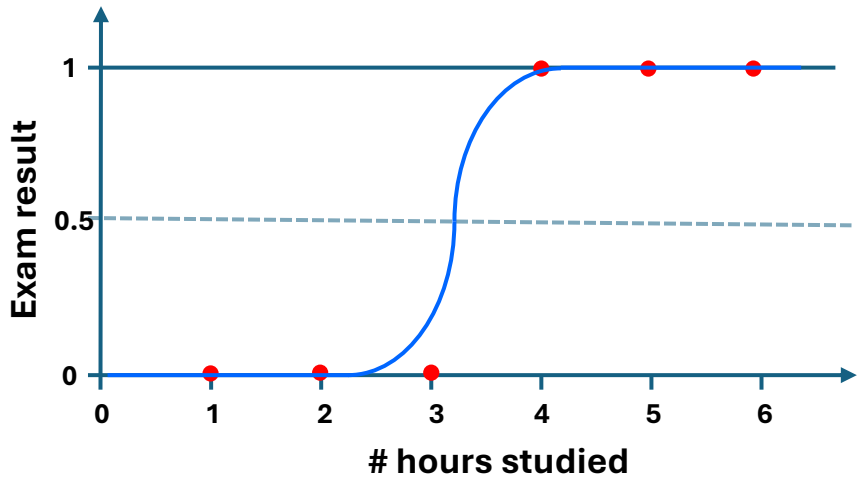
0=Fail
1=Pass

$$y = \frac{1}{1+e^{-(a_0+a_1x)}}$$

y = ?
a_o = -1.5
a_1 = 0.6
X = 3.3

$$y = \frac{1}{1+e^{-(-1.5+0.6 \text{ x } 3.3)}}$$

$$y = \frac{1}{1+e^{-(-0.48)}}$$



Classification: Logistic Regression

The Working of Logistic Regression

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**1**

0=Fail
1=Pass

A student who studied 3.3 hours has a 62% predicted probability of passing

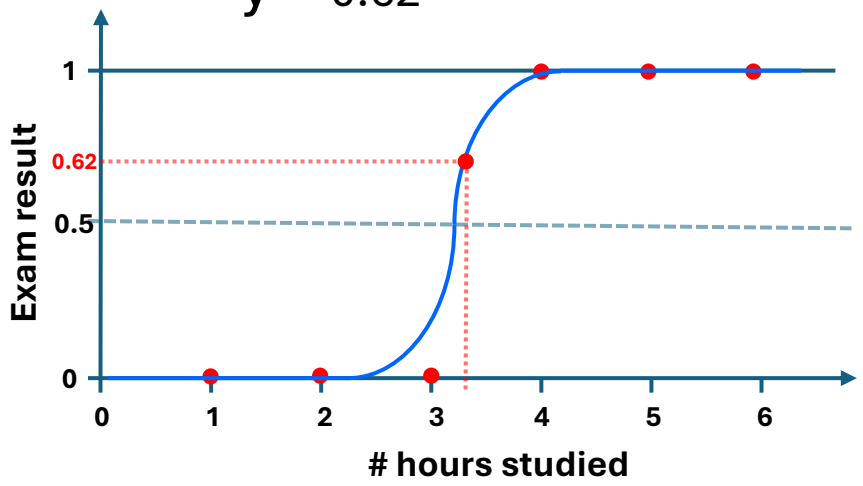
$$y = \frac{1}{1+e^{-(a_0+a_1x)}}$$

$$y = \frac{1}{1+e^{-(-1.5+0.6 \times 3.3)}}$$

$$y = \frac{1}{1+e^{-(-0.48)}}$$

$$y = 0.62$$

y = ?
a_o = -1.5
a_1 = 0.6
X = 3.3



Classification: Logistic Regression

The Working of Logistic Regression

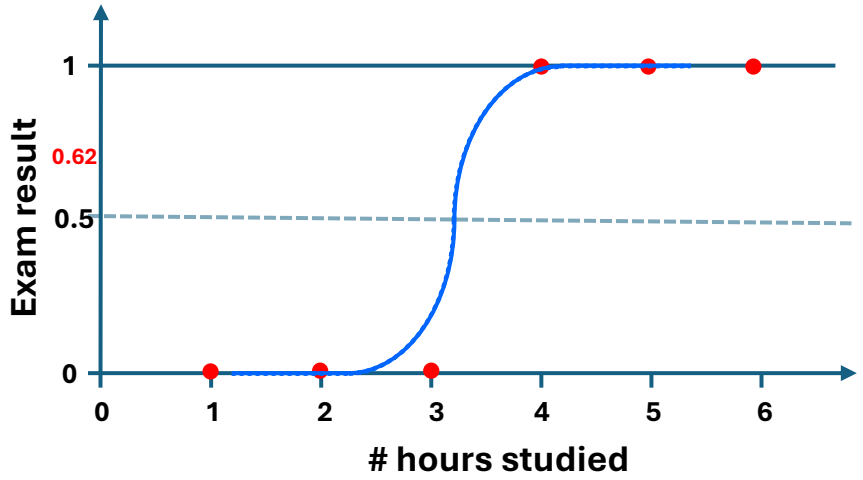
$$y = \frac{1}{1+e^{-(a_0+a_1x)}}$$

How to find a_0 *and* a_1*?*

- Different values of a_0 and a_1, are tried in training to checks how well each pair predicts the students whose results we already know.
- It keeps adjusting until it finds the pair that separates pass from fail best. Those two numbers are the model.

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**1**

0=Fail
1=Pass



Classification: Logistic Regression

The Working of Logistic Regression

$$y = \frac{1}{1+e^{-(a_0+a_1x)}} \qquad y = \frac{1}{1+e^{-(z)}}$$

$$z = a_0 + a_1x$$

$$z = a_0 + a_1(\text{hours_study})$$

Study Hours	Exam Results
2	0
5	1
4	1
1	0
6	1
3	0
3.3	**1**

0=Fail
1=Pass

Classification: Logistic Regression

The Working of Logistic Regression

$$y = \frac{1}{1+e^{-(a_0+a_1x)}} \qquad y = \frac{1}{1+e^{-(z)}}$$

Study Hours	Exam Results	Drink Cups
2	0	2
5	1	3
4	1	1
1	0	3
6	1	0
3	0	2
3.3	**1**	1

$$z = a_0 + a_1x$$

$$z = a_0 + a_1(\text{hours_study}) + a_2(\text{drink_cups})$$

Classification: Logistic Regression

The Working of Logistic Regression

$$y = \frac{1}{1+e^{-(a_0+a_1x)}} \qquad y = \frac{1}{1+e^{-(z)}}$$

Study Hours	Exam Results	Drink Cups	Extra Class
2	0	2	1
5	1	3	0
4	1	1	0
1	0	3	1
6	1	0	1
3	0	2	1
3.3	**1**	1	0

$$z = a_0 + a_1x$$

$$z = a_0 + a_1(\text{hours_study}) + a_2(\text{drink_cups}) + a_3(\text{extra_class})$$

Classification: Logistic Regression

Our Question For Today?

The scenario

A property assessor must place every house in a price band before the valuation is finalized.
They have the full record: size, quality, year built, garage, everything.
They do not yet have the price.

The question?

Given everything else about the house, is it **above** or **below** the market median?

How Logistic Regression asks it differently?

- k-NN asked: which houses are most similar to this one?
- Logistic Regression asks: what score do we get when we add up every column, and does that score land above or below the line?

Classification: Support Vector Machine (SVM)

Definition

A support vector machine finds the decision boundary that leaves the widest possible margin between the two classes, using only the training points closest to that boundary.

Chapter 4
Introduction to Data Mining, 2nd Edition

How it differs from logistic regression

- Logistic regression fits **all** the points.
- SVM looks only at the points **near the boundary**. Those are the support vectors.

Where to use

- Text and document classification, where columns outnumber rows
- Image classification, the standard method before deep learning
- Bioinformatics, thousands of genes and few samples
- Handwriting and face recognition

Classification: Support Vector Machine (SVM)

Types of SVM

Kernel	What it does	Use it when
Linear	A straight boundary	Many columns, or classes already separable
Polynomial	A curved boundary	The boundary bends in a regular way
RBF (radial basis)	A flexible, local boundary	The default. Works when you do not know the shape

How it works?

- Find the support vectors
- Add the bias(b) value in the vectors to convert 2D to 3D form.
- Consider parameter for every support vector like $\alpha 1, \beta 2 \;\; etc$
- Find the linear equations and solve them to find parameter values
- Final equation for hyperplane

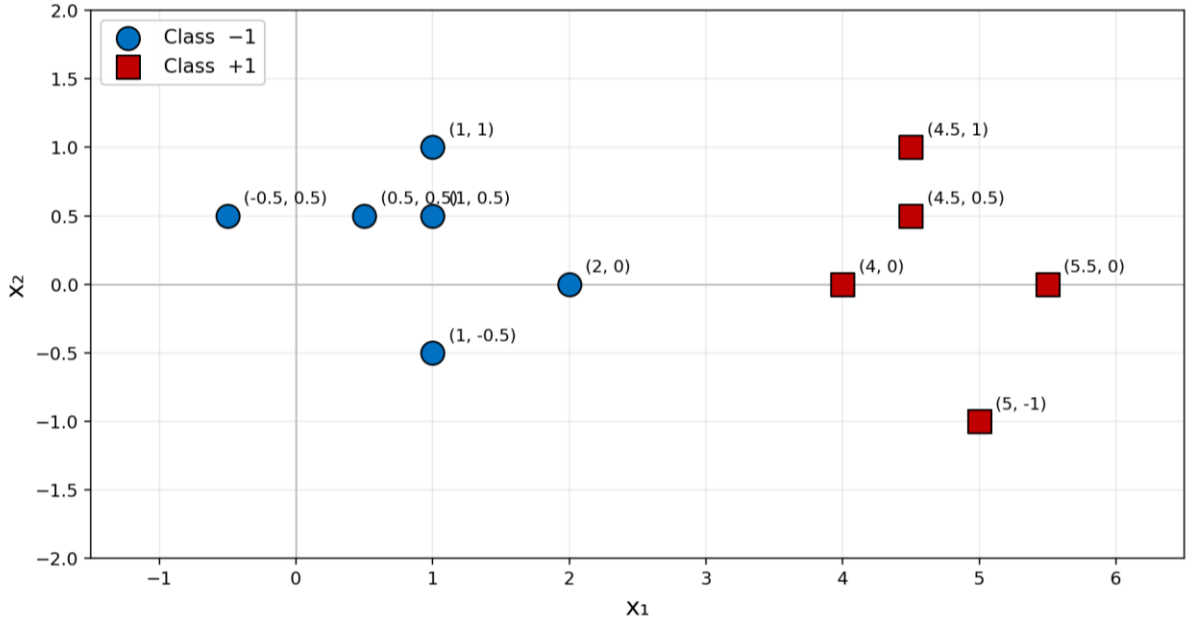
Classification: Support Vector Machine (SVM)

The Working of SVM

Classify animal into cats(-1) and dog (+1)

Data point (weight, height)

Class (-1): (1, 0.5), (1, 1), (1, -0.5), (-0.5, 0.5), (0.5, 0.5), (2, 0)
Class (+1): (4, 0), (4.5, 1), (4.5, .5), (5, -1), (5.5, 0)



Classification: Support Vector Machine (SVM)

The Working of SVM

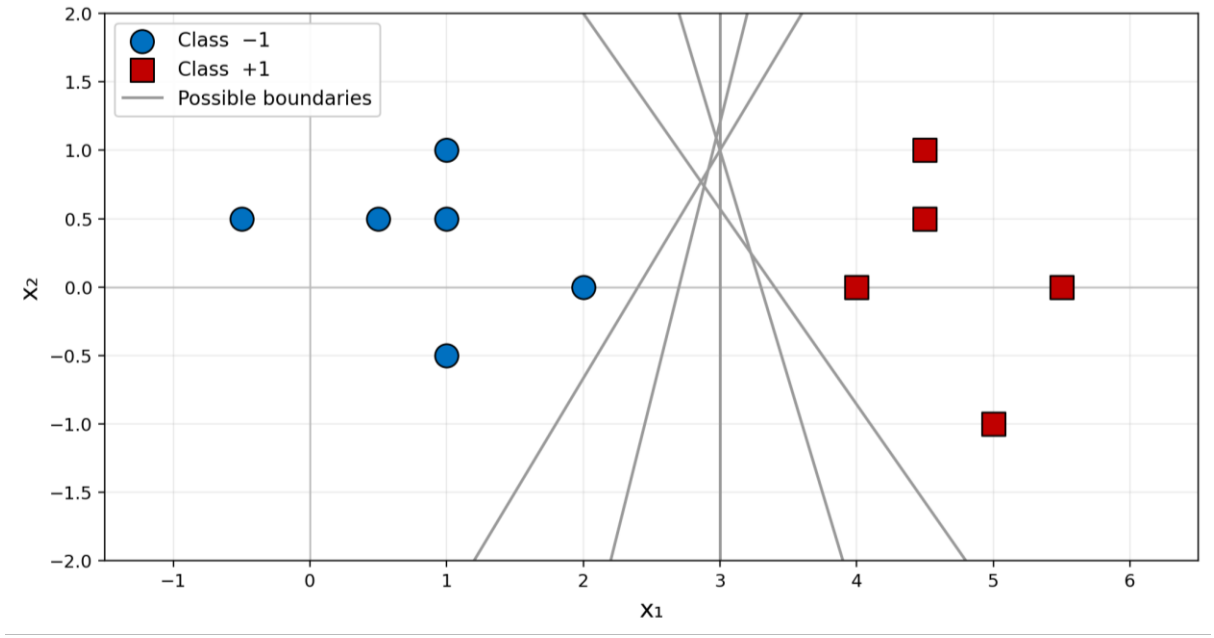
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Class (+1): (4, 0), (4.5, 1), (4.5, .5), (5, -1), (5.5, 0)

Hyperplane: A hyperplane is a decision boundary that separates data points of different classes in the feature space.

Mathematics representation: **w . x + b = 0**



Classification: Support Vector Machine (SVM)

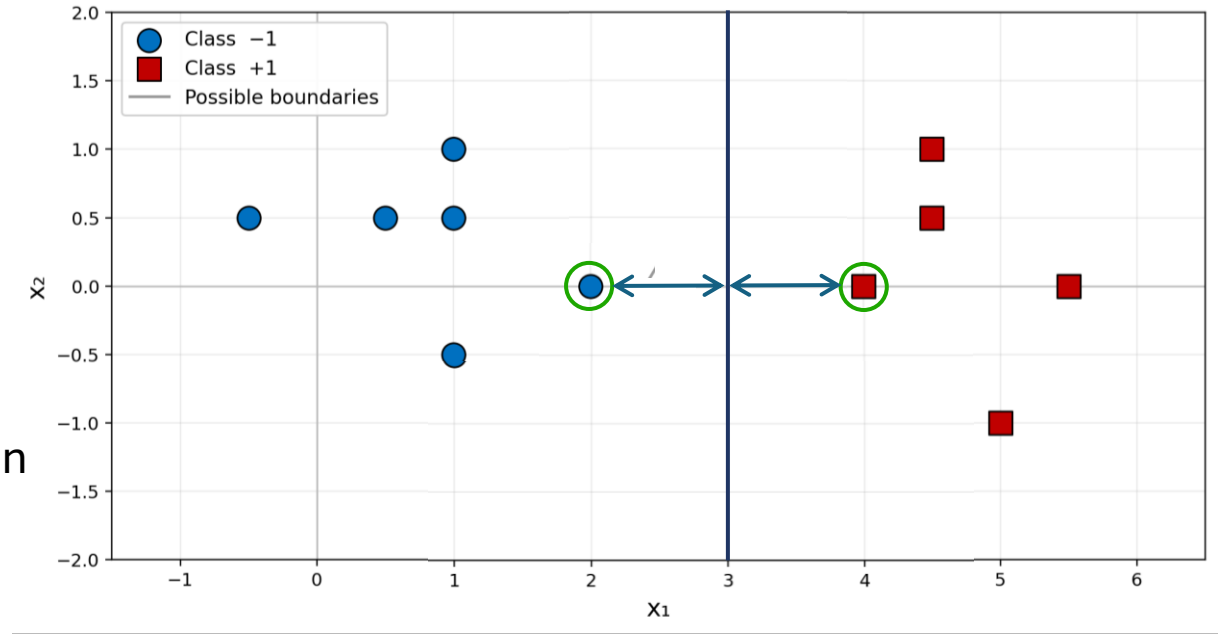
The Working of SVM

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Class (-1): (1, 0.5), (1, 1), (1, -0.5), (-0.5, 0.5), (0.5, 0.5), (2, 0)
Class (+1): (4, 0), (4.5, 1), (4.5, .5), (5, -1), (5.5, 0)

Support Vectors: These are the points that are closest to the hyperplane. These points influence the orientation and position of the hyperplane.



Classification: Support Vector Machine (SVM)

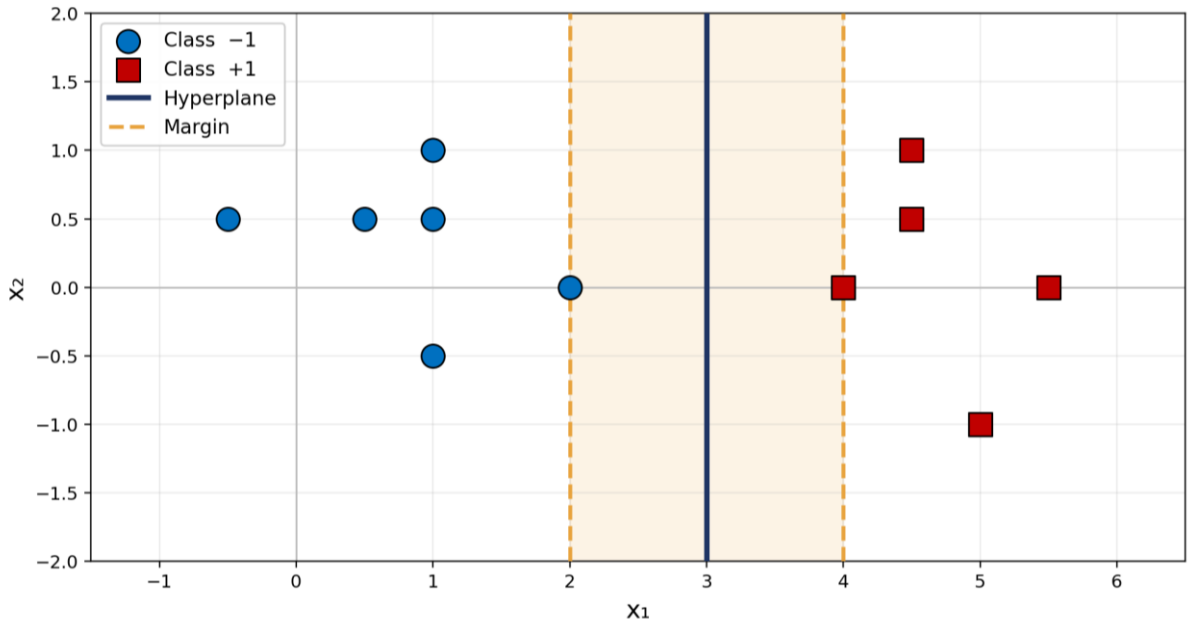
The Working of SVM

Classify animal into cats(-1) and dog (+1)

Data point (weight, height)

Class (-1): (1, 0.5), (1, 1), (1, -0.5), (-0.5, 0.5), (0.5, 0.5), (2, 0)
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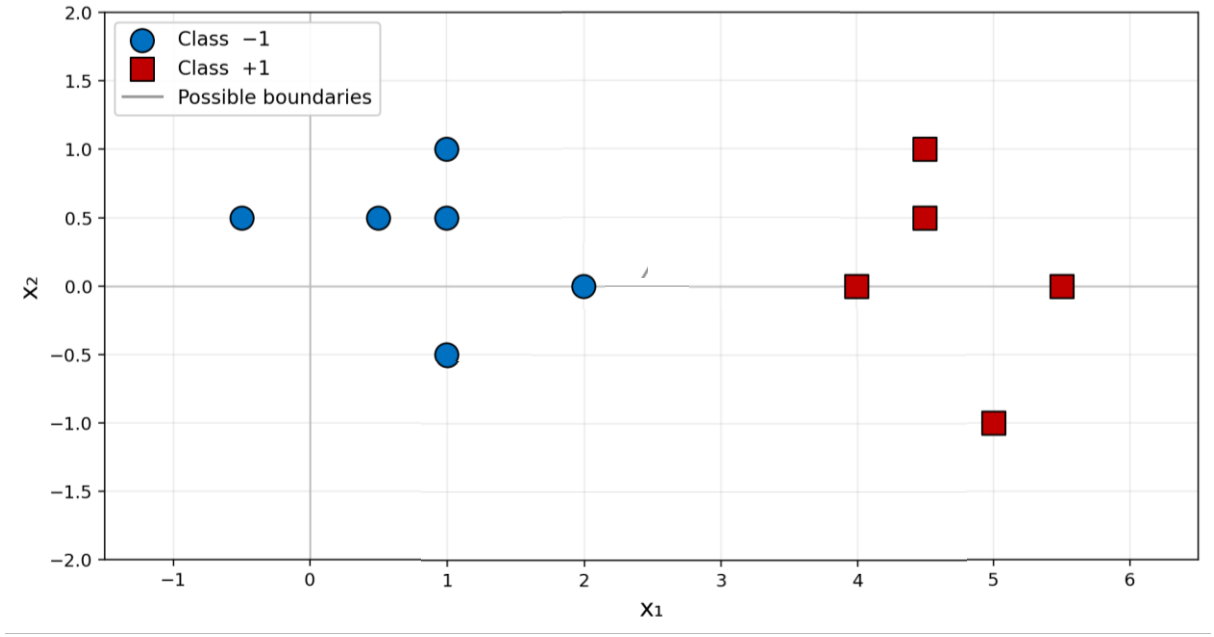
Margin: Margin is the distance between the hyperplane and the nearest data point of either class, SVM tries to maximize this margin to improve the classification confidence.



Classification: Support Vector Machine (SVM)

The Working of SVM

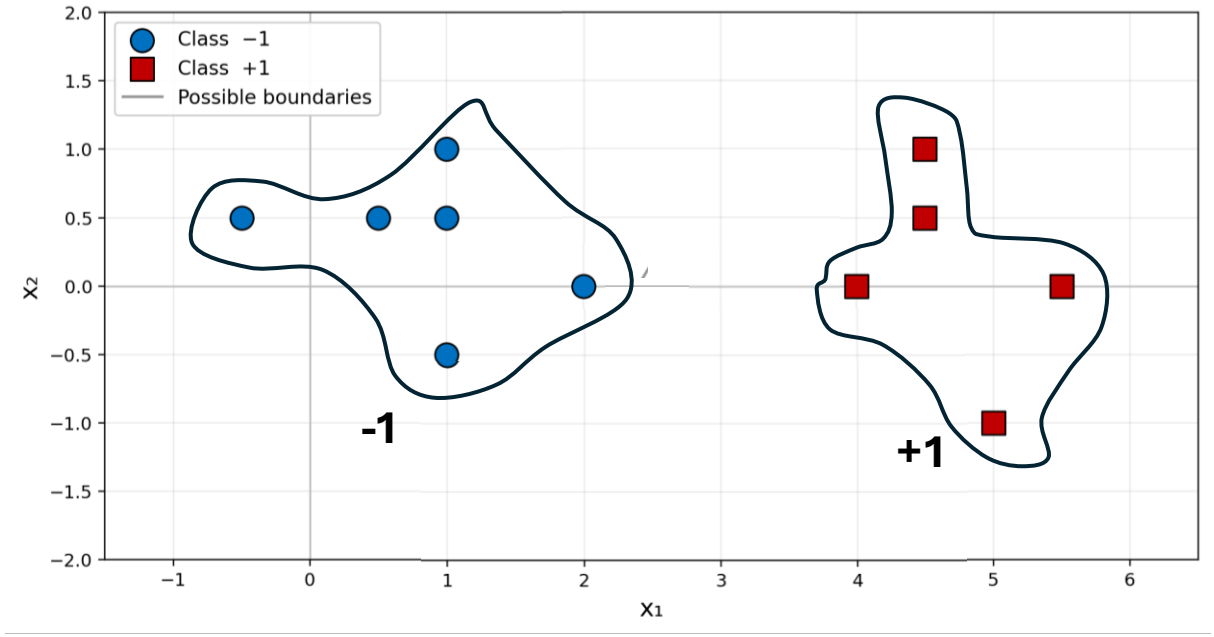
How to find the Hyperplane?



Classification: Support Vector Machine (SVM)

The Working of SVM

- Find the support vectors

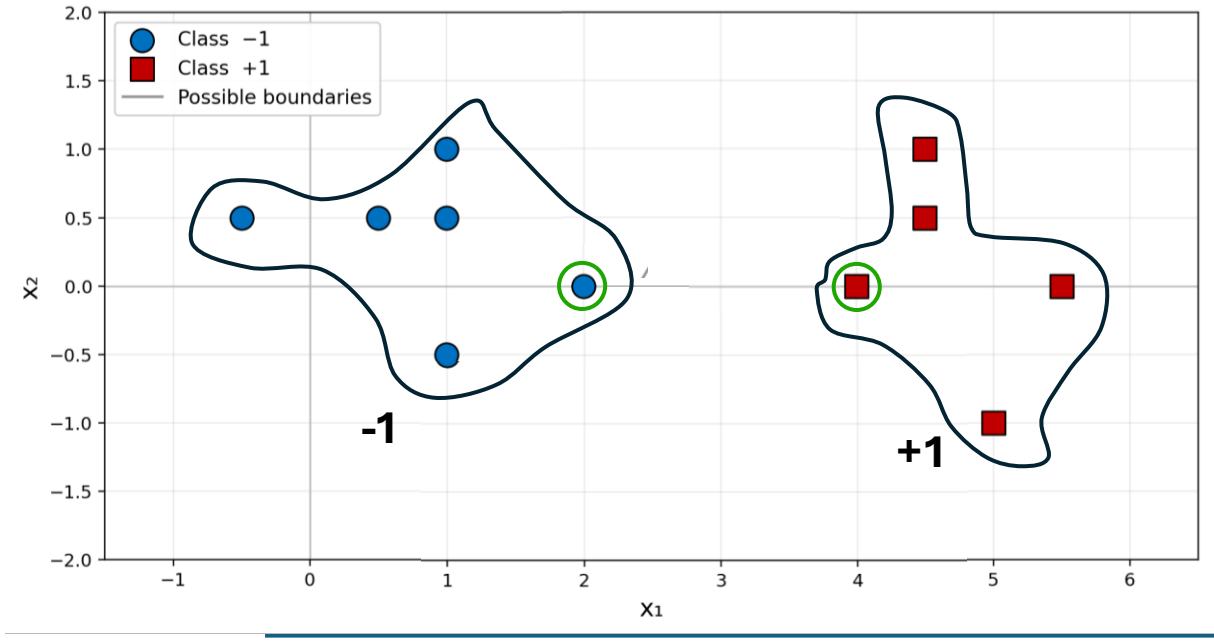


Classification: Support Vector Machine (SVM)

The Working of SVM

- Find the support vectors

$$S_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \qquad S_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

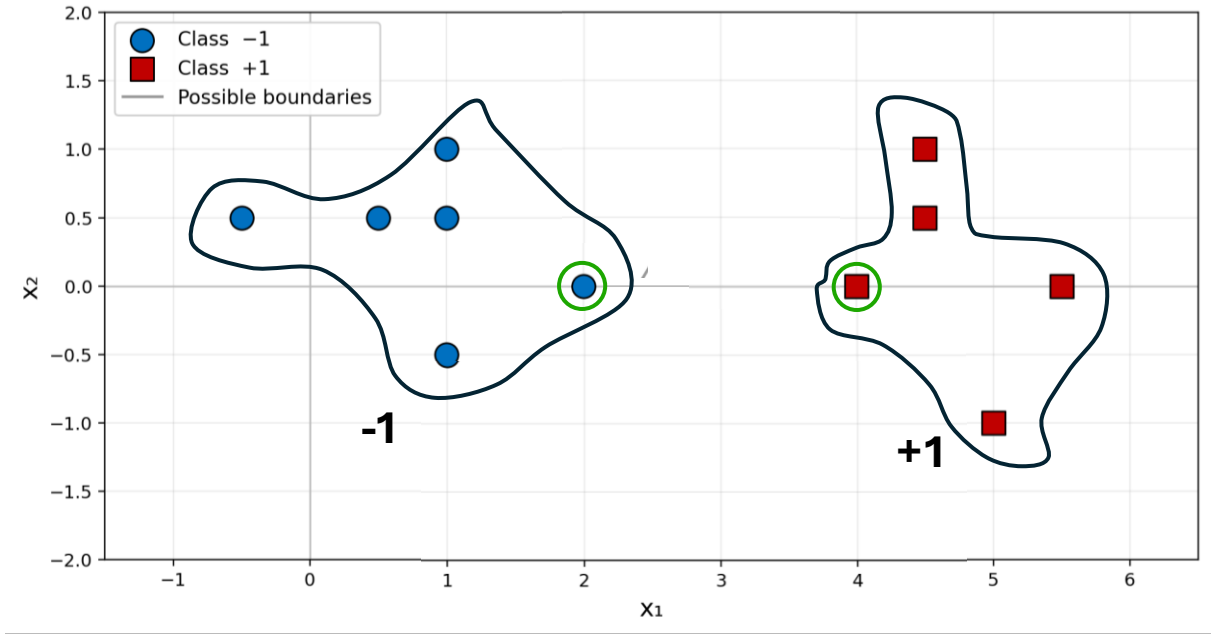


Classification: Support Vector Machine (SVM)

The Working of SVM

- Add the bias(b) value in the vectors to convert 2D to 3D form.

$$S_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \qquad S_2 = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$



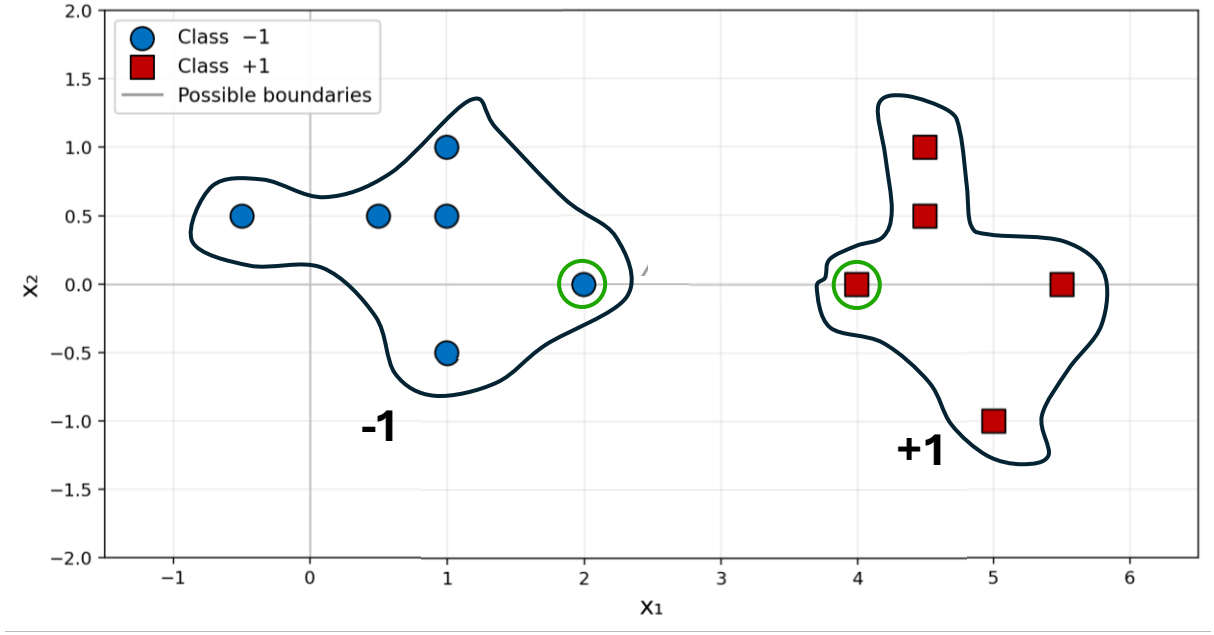
Classification: Support Vector Machine (SVM)

The Working of SVM

- Consider parameter for every support vector like α_1, β_2 etc

$$S_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \qquad S_2 = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

α_1 α_2



Classification: Support Vector Machine (SVM)

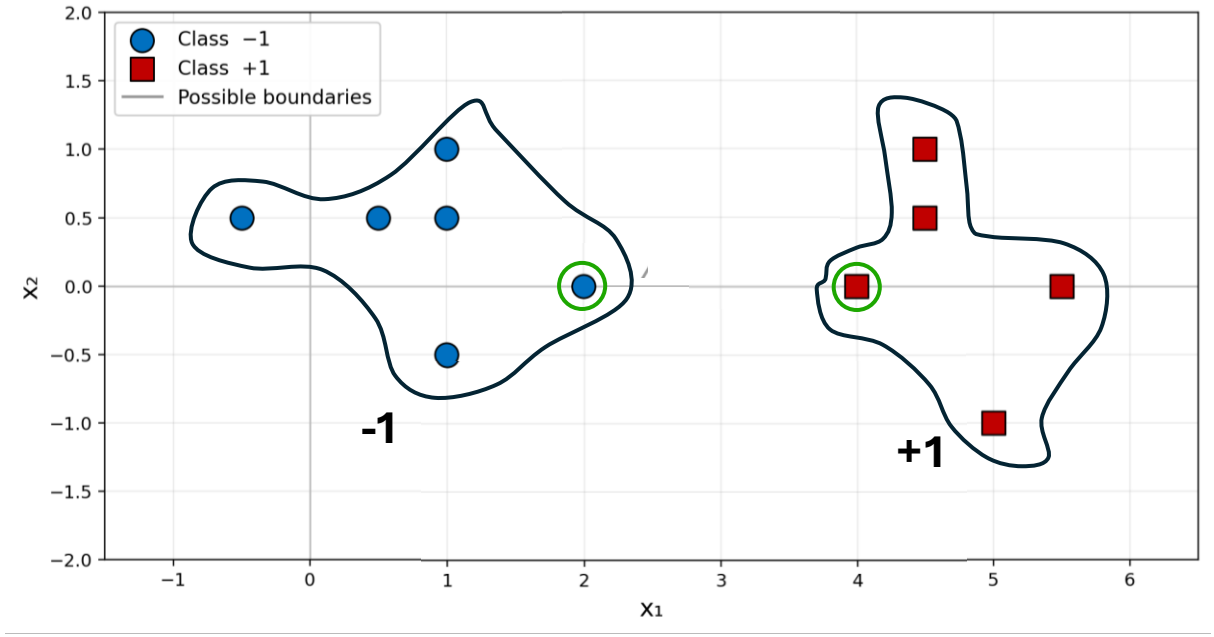
The Working of SVM

- Find the linear equations and solve them to find parameter values

$$S_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \qquad S_2 = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

$$\alpha_1 \qquad \alpha_2$$

$$\alpha_1.S_1 S_1 + \alpha_2.S_2 S_1 = -1 \qquad \alpha_1.S_1 S_2 + \alpha_2.S_2 S_2 = 1$$



Classification: Support Vector Machine (SVM)

The Working of SVM

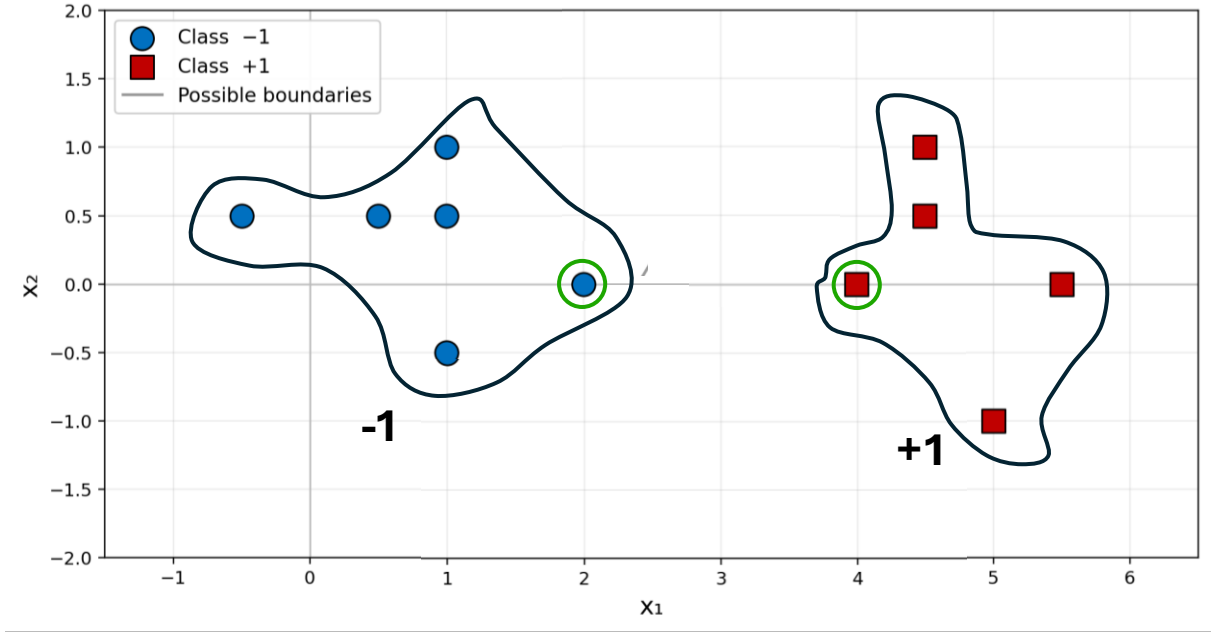
- Find the linear equations and solve them to find parameter values

$$S_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \qquad S_2 = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

$$\alpha_1 \qquad \alpha_2$$

$$\alpha_1.S_1S_1 + \alpha_2.S_2\ S_1 = -1 \qquad \alpha_1.S_1S_2 + \alpha_2.S_2\ S_2 = 1$$

$$\alpha_1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = -1 \qquad \alpha_1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} = 1$$



Classification: Support Vector Machine (SVM)

The Working of SVM

- Find the linear equations and solve them to find parameter values

$$S_1 = \begin{bmatrix}2\\0\\1\end{bmatrix} \qquad S_2 = \begin{bmatrix}4\\0\\1\end{bmatrix}$$

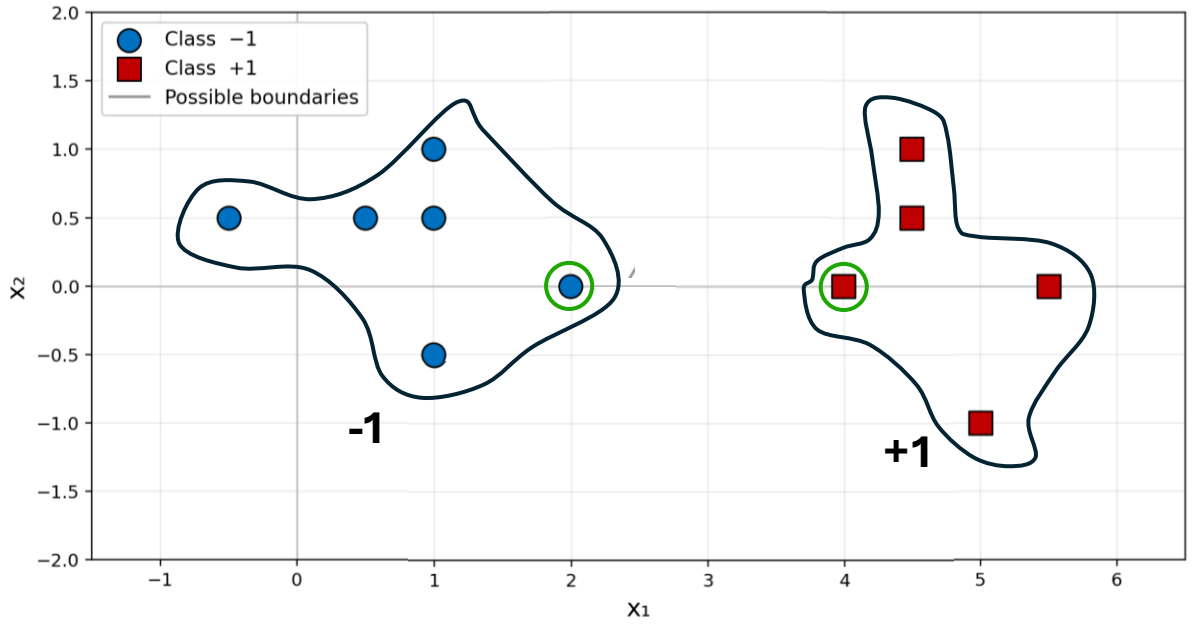
$$\alpha_1 \qquad \alpha_2$$

$$\alpha_1.S_1S_1 + \alpha_2.S_2\ S_1 = -1 \qquad \alpha_1.S_1S_2 + \alpha_2.S_2\ S_2 = 1$$

$$\alpha_1 \begin{bmatrix}2\\0\\1\end{bmatrix}\begin{bmatrix}2\\0\\1\end{bmatrix} + \alpha_2 \begin{bmatrix}4\\0\\1\end{bmatrix}\begin{bmatrix}2\\0\\1\end{bmatrix} = -1 \qquad \alpha_1 \begin{bmatrix}2\\0\\1\end{bmatrix}\begin{bmatrix}4\\0\\1\end{bmatrix} + \alpha_2 \begin{bmatrix}4\\0\\1\end{bmatrix}\begin{bmatrix}4\\0\\1\end{bmatrix} = 1$$

$\alpha_1 5 + \alpha_2 9 = -1$

$\alpha_1 9 + \alpha_2 17 = 1$



Classification: Support Vector Machine (SVM)

The Working of SVM

- Find the linear equations and solve them to find parameter values

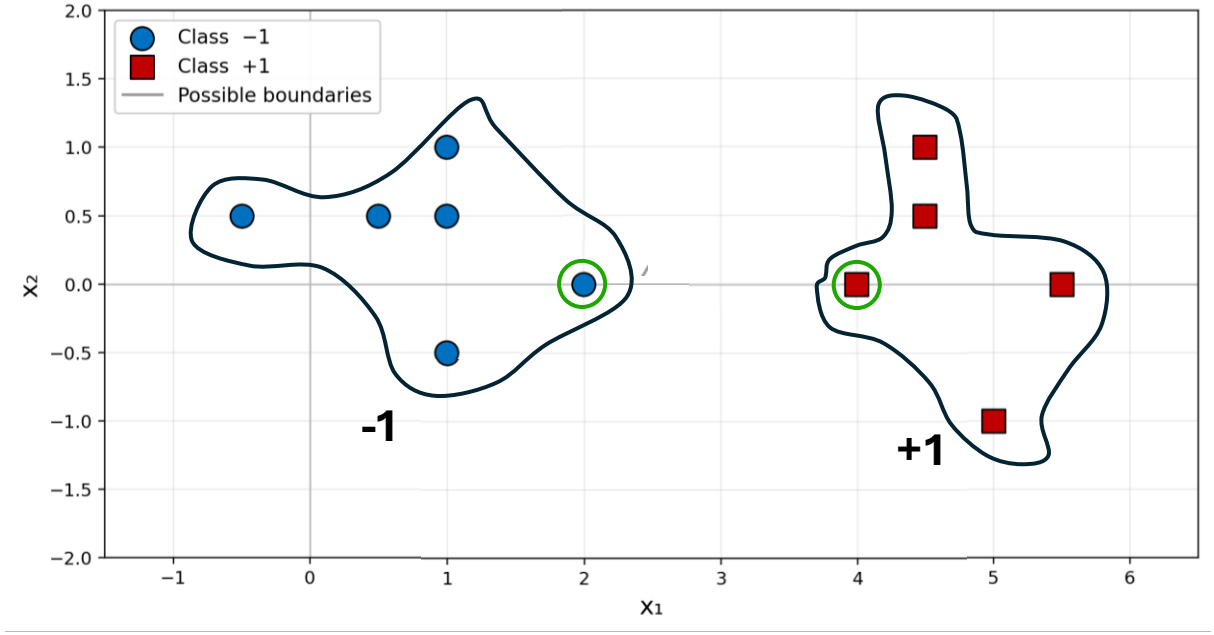
$\alpha_1 5 + \alpha_2 9 = -1$ eq(1)

$\alpha_1 9 + \alpha_2 17 = 1$ eq(2)

After solving these 2 linear equations we get:

$\alpha_1 = -6.5$

$\alpha_2 = 3.5$



Classification: Support Vector Machine (SVM)

The Working of SVM

- Final equation for hyperplane:

$$\overline{w} = \sum_{i=1}^{n} \alpha_i \, s_i$$

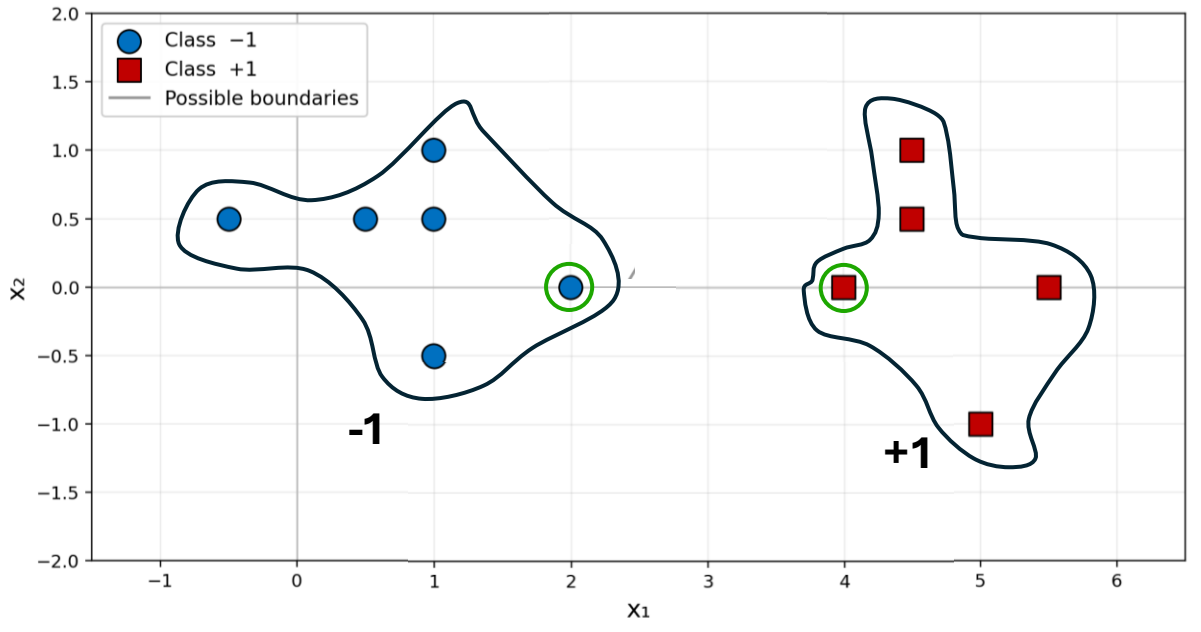
$$\overline{w} = \alpha_1 \, s_1 + \alpha_2 \, s_2$$

$$\overline{w} = -6.5 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + 3.5 \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

$$\overline{w} = \begin{bmatrix} -13 \\ 0 \\ -6.5 \end{bmatrix} + \begin{bmatrix} 14 \\ 0 \\ 3.5 \end{bmatrix} \qquad \overline{w} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$$

$\alpha_1 = -6.5$

$\alpha_2 = 3.5$



Classification: Support Vector Machine (SVM)

The Working of SVM

- Final equation for hyperplane:

$$\overline{w} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$$

Orientation of the line

Biased value

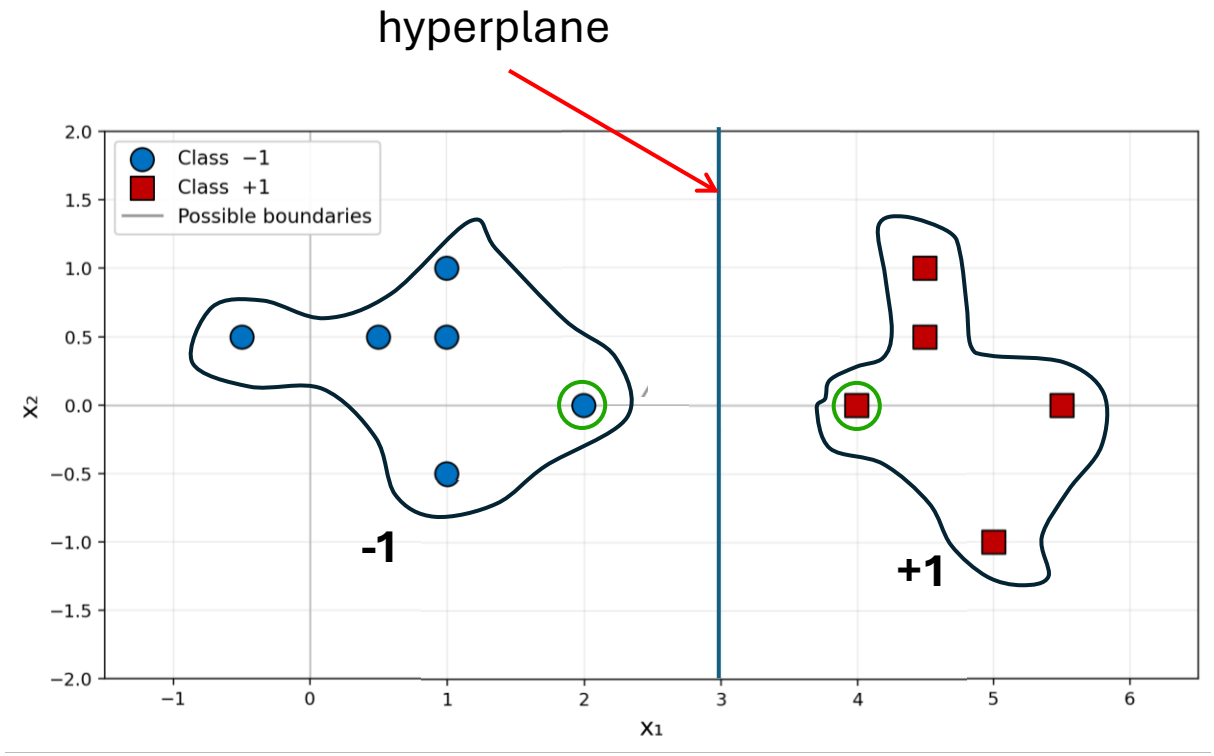
We added biased (b) in initial equation so,

b=-3 or b+3 = 0

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ Vertical line

$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ Horizontal line

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ 45 degree



Classification: Support Vector Machine (SVM)

The Working of SVM

- Final equation for hyperplane:

$$\overline{w} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$$

Orientation of the line

Biased value

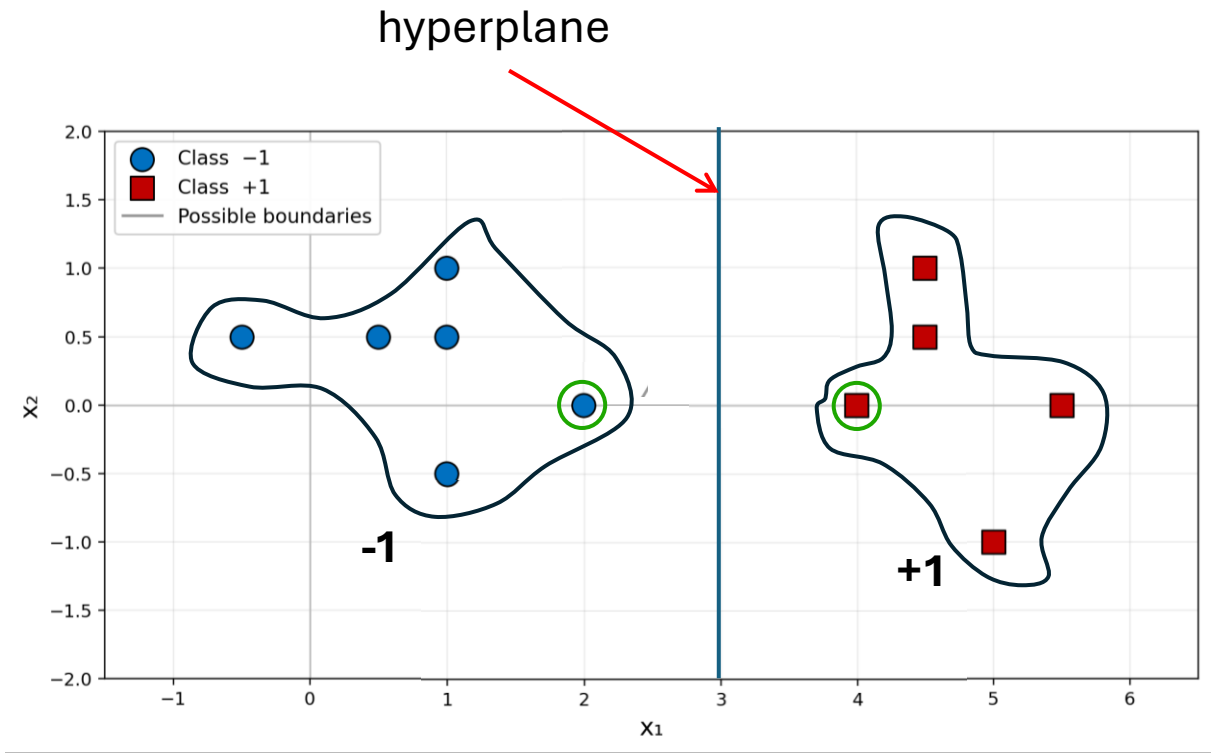
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Why Accuracy Is Not Enough

A Model That Scores 90 Percent and Finds Nothing

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

- **A new question for the same houses:**
  Not above or below median. Is this a LUXURY property? Top 10 percent of prices.
  293 luxury houses out of 2,930.

- **Now build the laziest possible model:**
  It predicts “not luxury” for every single house. It reads no columns at all.
  Accuracy: 90 percent.

The Confusion Matrix

Four Numbers Instead of One

	Predicted: NOT luxury	**Predicted: LUXURY**
Actually NOT luxury	True Negative	False Positive
Actually LUXURY	False Negative	True Positive

True Negative	an ordinary house, correctly called ordinary
True Positive	a luxury house, correctly spotted
False Positive	we flagged an ordinary house. The assessor wasted a site visit
False Negative	a luxury house we missed. It was undervalued and sold too cheap

Precision and Recall

Two Questions, Two Different Denominators

- **Precision asks: when we say luxury, how often are we right?**

```
Precision  =  True Positives  /  everything we CALLED luxury
```

$$Precision = \frac{TP}{TP + FP}$$

- **Recall asks: of all the real luxury houses, how many did we find?**

```
Recall     =  True Positives  /  everything that ACTUALLY is luxury
```

$$Recall = \frac{TP}{TP + FN}$$

- **The difference is what you divide by.**
  Precision divides by what the model claimed. Recall divides by what is out there.

Precision is about trusting the alarm. Recall is about not missing anything.

The Tradeoff

You Cannot Maximise Both

Threshold	What happens	Precision	Recall
0.9 very cautious	Only flags the obvious ones	High	Low
0.5 the default	Balanced	Medium	Medium
0.1 very eager	Flags anything remotely likely	Low	High

Push the threshold DOWN: you catch more real luxury houses, but you flag more ordinary ones too.
Push it UP and the reverse happens.

The threshold is a business decision, not a technical one. 0.5 is a default, not an answer.

F1, and When to Use It

One Number Again, But an Honest One

```
F1  =  2 x (Precision x Recall)  /  (Precision + Recall)
```

$$F1 = 2 \cdot \frac{precision \cdot recall}{precision + recall}$$

- **Why not just average them?**
  A plain average of 1.0 and 0.0 gives 0.5. That looks acceptable.
  F1 of 1.0 and 0.0 gives 0. Correctly brutal.
  F1 punishes you for being bad at either one.

- **Use F1 when the classes are imbalanced and both errors cost you something.**

- **Do not use F1 when one error is far worse. Then optimise that one directly.**

Which Error Is Worse?

Three Situations. Answer Before I Do.

- **Cancer screening.**
  False positive: a healthy patient gets a frightening follow-up test.
  False negative: a tumour goes undetected.
- **Spam filter.**
  False positive: a real email lands in the spam folder and is never read.
  False negative: one spam message reaches the inbox.
- **Our luxury property question.**
  False positive: a wasted site visit. False negative: a house sold too cheap.

- **Different answers. So a different metric to optimise in each case.**

ROC and AUC

Every Threshold at Once, in One Picture

- **Slide the threshold from 1 down to 0 and plot two things:**
  Up the y axis: True Positive Rate. The same thing as recall.
  Along the x axis: False Positive Rate. How many ordinary houses you wrongly flagged.

- **AUC is the area under that curve.**
  1.0 is perfect. 0.5 is a coin flip. Below 0.5 means something is wired backwards.

- **AUC is threshold free. It judges the ranking, not one cut-off.**

Cross-Validation

Five Splits, Not One

- One split gives you one number, and that number got lucky or unlucky.
- Change the random seed and it moves. That is not evidence.

- **So do it five times:**

  Split the training data into 5 parts. Hold out part 1, train on the rest, score it.
  Then hold out part 2. And so on. Five scores. Average them.

- **Report the standard deviation, not just the mean.**
- If two models differ by 0.5 points and the spread is 2.0, they are tied.

- **Put the scaler INSIDE the pipeline, or each fold sees its own test data.**

Leakage

The Fastest Way to Fool Yourself

- Leakage is when information from the test set reaches the model during training.
- The score comes out excellent. It means nothing.

- **Four ways it happens:**
  A column derived from the target. Predicting price with price per square foot in X.
  Scaling or imputing before the split, so the test set shaped the training set.
  Tuning until the test score looks good. You fitted to the test set.
  Time travel. Using information that did not exist when the prediction was made.

- **How to catch it: if your accuracy is suspiciously high, look for it. Do not celebrate.**